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KRONUMOS AI (TOKENECTOMY LABS)
EXECUTIVE TECHNICAL & COMMERCIAL DOSSIER – INTERNATIONAL ENGLISH EDITION
HIGH-CONVICTION PROPOSAL FOR AI ACCELERATORS, GRANTS & ENTERPRISE PARTNERS

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1. EXECUTIVE SUMMARY & CORPORATE IDENTITY

- * Entity & Trade Name:
Kronumos AI (Tokenectomy Labs)
- * Primary Research & Product Showcase:
<https://huggingface.co/Kronumos>
- * Open-Source Repository (GitHub):
<https://github.com/Tokenectomy-Labs/Kronomus>
- * Verified Academic Provenance:
Springer Nature Portfolio (Research Square):
<https://doi.org/10.21203/rs.3.rs-11205335/v1>
CERN Zenodo Permanent Scientific Archive:
<https://doi.org/10.5281/zenodo.22929676>
- * Lead Author & Research Anchor:
Muhammad Naufal Daffa – CTO & Chief Scientist
ORCID: <https://orcid.org/0009-0000-7909-4916>
- * Operational Headquarters:
Indonesia (Delaware C-Corp Structure via Stripe Atlas in Progress)
- * Capitalization Stage:
Bootstrapped / Pre-Seed (Empirically Validated Working MVP)

2. THE STRATEGIC PROBLEM: THE CONTEXT BLOAT & FRAGILITY WALL

Enterprise software engineering organizations face prohibitive compute costs and severe syntactic brittleness when deploying AI coding agents for automated software repair. Monolithic LLM agent architectures (such as Devin, Claude Engineer, or generic SWE-agents) treat debug logs as raw unstructured text, stuffing 50,000 to 150,000 tokens of noisy traceback frames and third-party library dependencies into commercial model context windows.

This creates three critical industry failure modes:

1. Ruinous Unit Economics:
Remediating a single GitHub issue costs \$2.00 to \$5.00 on commercial frontier APIs (reaching \$15.00 on multi-turn failure loops). For an enterprise engineering team processing 500 bugs monthly, token costs alone exceed \$2,500/month.
2. Syntactic Hallucination & CI Breakage:
Because autoregressive language models predict tokens statistically, they frequently hallucinate whitespace, tab-space indentation, and diff header offsets. Over 60% of candidate patches are rejected outright by 'git apply', breaking production CI/CD.

3. Data Sovereignty & IP Vulnerability:

Regulated industries (banking, fintech, defense, and healthcare) are legally barred by compliance mandates (SOC2, ISO 27001, HIPAA) from transmitting internal proprietary source code to multi-tenant, closed-source cloud APIs.

3. THE SOLUTION: DUAL-BRAIN NEURO-SYMBOLIC APR ARCHITECTURE

Kronumos resolves this architectural bottleneck through a cybernetic Dual-Brain design that enforces a strict separation between semantic reasoning and deterministic compiler mechanics:

1. Deterministic Rust Sub-Cortex Kernel (libtokenectomy_subcortex.so, C-ABI 5µs):

A zero-allocation native compiler kernel that executes before neural inference. It

parses stack traces at the byte level, deterministically excises 93.5% of framework

noise (node_modules, site-packages, stdlib), and extracts the precise AST function slice.

Following candidate code generation, native Rust modules (IndentationHealer and ScopeGuard)

reconcile block whitespace and verify variable bindings in sub-microsecond latency

without consuming model tokens.

2. Specialized Neural Cortex Models:

A tiered neural reasoning stack comprising:

* Kronumos Kairos (7B/14B Qwen-based fine-tuned models for rapid edge/CI execution)

* Kronumos Aion (671B MoE DeepSeek-V3 based architecture for complex, multi-file enterprise refactoring).

Empirical Outcome:

Context windows shrink from 150,000 tokens to an average of ~2,500 surgical tokens. Per-patch remediation costs drop below \$0.05, while delivering 100% clean POSIX unified

diffs that apply cleanly on production branches (Zero Dirty Diffs).

4. THE 5 CORE STRATEGIC DIFFERENTIATORS

1. Extreme Cost Efficiency (93.5% Context Reduction / Cost-Bounded Economics):

Sub-Cortex deterministic pruning excises external trace frames in 5µs, dropping prompt

bloat from 150,000 tokens to ~2,500 tokens and slashing inference costs from \$2-\$5 down

to <\$0.05 per patch.

2. Deterministic Syntax Integrity & Zero Dirty Diffs (Pre-Cognitive AST Healing):

Native Tree-sitter AST validation and indentation reconciliation enforce 100% POSIX

patch compliance before committing to git branches, eliminating CI build-breaker loops.

3. Dual-Brain Neuro-Symbolic Synergy:

Neural models focus solely on semantic reasoning while the native Rust kernel enforces syntax, memory limits, and scope invariants deterministically.

4. Enterprise Sovereign & Air-Gapped Deployment:

Deployable inside isolated private cloud environments (Azure VNet, On-Premise) with zero egress to third-party APIs, satisfying strict banking and defense compliance requirements.

5. Peer-Reviewed Scientific Provenance:

Rigorously benchmarked across 500 Princeton SWE-bench Verified instances, published on Springer Nature Research Square (DOI: 10.21203/rs.3.rs-11205335/v1), and permanently archived on CERN Zenodo (DOI: 10.5281/zenodo.22929676).

5. MARKET OPPORTUNITY, GTM & ENTERPRISE UNIT ECONOMICS

Target Customer Segments:

- * Enterprise Engineering Orgs & B2B SaaS: Teams with 50-1,000+ developers automating regression triage across sprint cycles.
- * Highly Regulated Sectors: Banking, fintech, defense, and healthcare requiring on-premise APR without source code egress.
- * DevTool & CI/CD Platform Providers: Platforms integrating a headless, low-latency code repair engine into their build pipelines.

Commercial Tiering Structure:

- * Community Open-Source Tier (Dual License / Apache 2.0): Open-weight models (Kronumos Kairos) and CLI for grassroots global developer adoption.
- * Developer Pro & Team Cloud (\$49 - \$199 / month per team): Managed cloud API for automated GitHub Actions bots and IDE extensions (VS Code / Cursor).
- * Enterprise Sovereign License (\$40,000 - \$80,000 / year per license): Air-gapped self-hosted binary on Azure VNet or on-premise clusters, unlimited nodes, custom AST rules, and 24/7 dedicated enterprise SLA.

Enterprise Customer ROI Calculation:

- * Baseline: A 100-developer organization resolving 500 issues/month spends ~\$1,500/month on raw LLM tokens plus ~150 hours of developer time fixing rejected diffs (~\$11,250 in developer salary). Total monthly cost: ~\$12,750.
- * Kronumos Implementation: 500 issues x 2,500 tokens = \$18.75/month in compute tokens. Zero dirty diffs eliminates manual patch fixing. Net enterprise savings exceed \$10,000/month, yielding full ROI within weeks.

6. CLOUD COMPUTE RESOURCE UTILIZATION PLAN (AZURE INFRASTRUCTURE)

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- * Stage 1: Initial Inception & Standardized Endpoints (\$1,000 - \$5,000):
Deploy single-GPU Azure NC A100 v4 instances to host standardized OpenAI-compatible inference endpoints for Kronumos Kairos, alongside encrypted Azure Blob Storage for benchmark artifact logging.
 - * Stage 2: Mass Model Serving & Container Sandboxing (\$25,000):
Provision multi-GPU Azure ND A100 v4 clusters (8x A100 80GB SXM4) for high-throughput vLLM serving of Kronumos Aion (671B MoE), coupled with Azure Kubernetes Service (AKS) clusters running 500 parallel isolated Docker evaluation harnesses.
 - * Stage 3: Commercial Scaling & GitHub Actions Bot Fleet (\$150,000):
Scale multi-GPU Azure ND H100 v5 compute nodes to support concurrent enterprise pilot workloads with sub-second latencies; launch automated GitHub Actions remediation bot on the Microsoft GitHub Marketplace.

7. EXECUTIVE FOUNDING ROSTER & DOMAIN AUTONOMY

1. Muhammad Naufal Daffa – Chief Technology Officer (CTO) & Chief Scientist
Domain: Core AI/ML Systems, Native Rust Compiler Architecture, Academic Research.
Credentials: First author of the Springer Nature Research Square preprint; architect of libtokenectomy_subcortex.so (C-ABI 5µs); designer of Kronumos Kairos (7B/14B) and Kronumos Aion (671B MoE).
ORCID: <https://orcid.org/0009-0000-7909-4916>
2. Muhamad Sufyan – Chief Executive Officer (CEO) & Head of Business Operations
Domain: Commercial Strategy, B2B Enterprise Partnerships, Investor Relations, Legal & Compliance.
Scope: Overseeing cloud compute grant execution, legal incorporation (Delaware C-Corp via Stripe Atlas), enterprise pilot acquisition, and commercial SaaS/Sovereign licensing.
3. Adi Supriyadi – Head of Product & Lead Frontend Architect
Domain: Developer Experience (DX), Web Architecture, Client Monitoring Dashboard, Visual Tooling.
Scope: Designing real-time token reduction telemetry dashboards, official web portal, IDE extensions (VS Code / Cursor), and GitHub Actions user workflow integrations.

8. VERIFICATION LINKS & SCIENTIFIC CITATIONS

- * Springer Nature Research Square Preprint:
<https://doi.org/10.21203/rs.3.rs-11205335/v1>
"Kronumos 2 Kairos: Cost-Bounded Automated Program Repair via Dual-Brain Cybernetic Sub-Cortex on SWE-bench Verified"

* CERN Zenodo Permanent Scientific Archive:

<https://doi.org/10.5281/zenodo.22929676>

* Official GitHub Repository:

<https://github.com/Tokenectomy-Labs/Kronomos>

* Official Hugging Face Organization:

<https://huggingface.co/Kronomos>

Community Traction: 1,260+ organic model downloads.

* Hacker News Global Community Discussion:

"Show HN: Kronomos-Neurosymbolic code repair with a 5 μ s Rust Sub-Cortex"

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